TapesVRy: Immersive Panoramic Exploration in Large-Scale Video Retrieval

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Abstract. Efficient video discovery and retrieval remains a significant challenge, especially in massive video archives that can contain thousands of hours of content. Traditional browsing methods, such as timeline previews or metadata-based searches, only allow users to access content sequentially. This makes it difficult to get a comprehensive view or quickly identify the required content. Inspired by the concept of Video Tapestry, we present TapesVRy, a system that enables interactive, immersive video discovery. Instead of presenting frames in a flat layout, our system arranges and blends keyframes and short clips into a 360° panoramic tapestry that users can explore with Meta Quest VR headsets. Videos are automatically clustered by semantic similarity using multimodal AI models, grouping related videos into thematic "universes". In each universe, users can freely navigate the 360° space, zooming in or out to adjust the density of the content. By directly interacting with the image areas, they can immediately access the original video segments. This approach transforms passive video viewing into an active exploration process, allowing users to browse and intuitively grasp multiple videos in a short time.

Keywords: video browser showdown · retrieval systems · virtual reality · semantic embedding · video tapestry

1 Introduction

The rapid growth of online video platforms has led to an urgent need for more efficient ways to discover and search for content. With data stores that can reach tens of thousands of videos, traditional browsing methods such as keyword searching, time-based filtering, or thumbnail viewing have gradually shown their

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limitations: users have to browse sequentially, struggle to get an overview, and spend a long time finding the right content.

The Video Browser Showdown (VBS) competition [13–16] has become the benchmark environment for evaluating advanced video retrieval systems. Here, systems must support both targeted and undirected searches, with the requirement for fast, accurate retrieval of large video stores under time constraints. In addition to accuracy, systems also need to focus on ease of use, interoperability, and discovery support to enhance user experience in competitive environments.

We introduce **TapesVRy**, a system that reimagines video browsing as an immersive 360° exploration task. Instead of presenting videos through flat thumbnails or keyframe strips, TapesVRy arranges and blends representative frames into a panoramic tapestry that surrounds the user in virtual reality. With a Meta Quest headset, users can rotate, zoom, and directly interact with regions of this tapestry to access the underlying video content. To organize such large collections, TapesVRy uses multimodal AI models to semantically cluster videos into thematic *universes* (e.g., *indoor scenes*, *international events*). Each universe provides a contextually relevant space for exploring related videos intuitively.

Previous research on video browsing and summarization has proposed many solutions, such as storyboarding, keyframe merging, timeline previewing, or automatic video surfing [4]. A typical example is Video Tapestry [2], which visualizes the entire video as a compact collage of representative frames. However, this flat, two-dimensional approach only works for individual videos; when applied to thousands of videos, the layout becomes cluttered and less suitable for large-scale browsing tasks such as in VBS. In contrast, TapesVRy exploits an immersive VR environment to present content in a panoramic and open space, allowing users to both get an overview of the collection and easily drill down to specific segments. Furthermore, applying semantic clustering helps group videos by meaning rather than in arbitrary order, thereby supporting intuitive navigation and contextualizing the experience. This combination overcomes the limitations of traditional 2D methods and better meets the need for fast, accurate discovery in the competitive VBS landscape.

The structure of this paper is arranged as follows. In Sect. 2, we discuss the existing methods related to our research. Our proposed method will be detailed in Sect. 3. Specifically, Sect. 3.1 covers Video Preprocessing and Keyframe Extraction, while Sect. 3.2 addresses Semantic Embedding and Clustering. Additionally, Sect. 3.3 will introduce 360° Tapestry Generation, followed by Sect. 3.4, which focuses on Interactive VR-based Exploration and Retrieval. Lastly, Sect. 4 provides the conclusion.

2 Related Work

2.1 Video Summarization and Browsing

Video summarization is a popular research direction to help users navigate large video collections more efficiently. Typical techniques include static keyframe extraction [12], storyboarding [3], and dynamic video skimming [1]. These methods provide an overview but are largely based on linear sequencing and limited to two-dimensional layouts, making it difficult to represent relationships among multiple videos. Video Tapestry [2] proposed a way to visualize an entire video as a compact collage, serving as an alternative to storyboards. However, this flat design cannot scale to thousands of videos, as the mosaic layout easily becomes cluttered and visually overwhelming.

2.2 Video Retrieval and the Video Browser Showdown

The Video Browser Showdown (VBS) [13–16] competition has become a benchmark for evaluating video retrieval systems. Competing systems often incorporate multimodal search (text, sketch, video clip queries) [24] and efficient indexing strategies to support retrieval under strict time constraints. Interaction paradigms in VBS systems typically emphasize search bars, filtering interfaces, and ranked result lists, occasionally extended with grid-based keyframe browsers [8,22]. While effective for query-driven retrieval, these approaches rarely support exploratory browsing or provide intuitive overviews of large-scale collections.

2.3 Immersive and Interactive Video Exploration

Recent works in immersive analytics and VR-based video interaction [11,23] highlight the potential of spatial environments for exploring complex data. Systems leveraging 360° video or panoramic visualization have demonstrated advantages in providing spatial context and intuitive navigation [5]. However, most immersive approaches focus on single-video experiences (e.g., 360° films, VR cinema) rather than multi-video summarization and retrieval.

We hypothesize that extending tapestry-style summarization into a 360° interactive environment allows users to perceive video relations more naturally and reduces clutter, thereby improving observation and search efficiency in large-scale tasks such as VBS.

2.4 VR-Based Systems for the Video Browser Showdown

Virtual Reality (VR) has also been explored in the context of the VBS through the vitrivr-VR system, which has been continuously developed over several editions. Early versions introduced VR as a modality for multimedia retrieval, enabling query formulation via speech-to-text and result presentation in immersive environments [18]. Later releases added visual-text co-embedding in a shared latent embedding space and novel interfaces such as pose and temporal queries [17], improved text input methods and support for asynchronous workflows [19], and a point cloud browsing interface [20]. The most recent version integrates a more performant retrieval backend with modern VR hardware, such as the Apple Vision Pro, to enhance retrieval efficiency and input capabilities [21]. These developments illustrate the growing role of VR as a promising interface for video retrieval in VBS scenarios.

3 System Overview

Our system connects existing research on video summarization and immersive analytics. First, raw videos are preprocessed and representative keyframes are extracted. Second, multimodal embeddings are computed and clustered to group videos into semantic universes. Third, these clusters are converted into 360° images, with keyframes spatially arranged to reflect thematic relationships. Finally, users explore these immersive environments in VR, interacting with regions of the 360° images to retrieve and view the original video segments (Fig. 1).

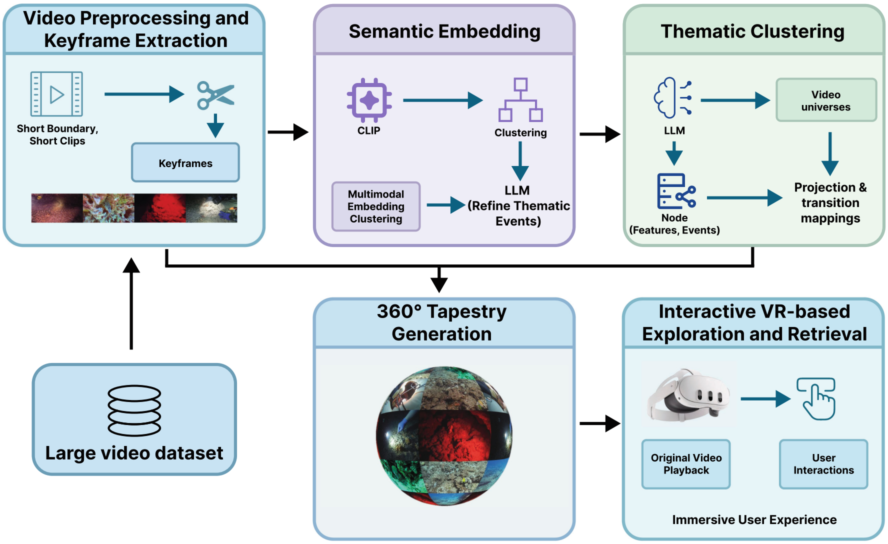

Fig. 1. System overview of TapesVRy. The workflow consists of four main components: (i) *Video Preprocessing and Keyframe Extraction* – sampling representative frames from large video collections, (ii) *Semantic Embedding and Thematic Clustering* – grouping frames into thematic universes using multimodal AI, (iii) *Panoramic Tapestry Generation* – stitching clustered frames into navigable 360° environments, and (iv) *Interactive VR-based Exploration and Retrieval* – enabling immersive browsing, zoom control, and direct video access via a VR headset.

3.1 Video Preprocessing and Keyframe Extraction

For each video in the dataset, we first extract a compact yet representative set of visual samples that capture the essence of the content. Instead of retaining every frame, which would introduce substantial redundancy and hinder scalability, we focus on identifying moments that best reflect the visual and semantic diversity within the video. Following standard practices in video summarization [7], our pipeline applies shot boundary detection to segment the video into coherent units. From each segment, we further analyze scene-level feature diversity and select 3–5 keyframes that collectively represent the video's distinctive elements, such as different objects, backgrounds, or activities.

In addition to static frames, we optionally extract short 2–3-second clips to preserve motion cues and temporal continuity, enriching the dynamic tapestry view used in later stages. This preprocessing step not only reduces redundant information but also establishes a concise and structured visual foundation that facilitates subsequent semantic embedding, clustering, and immersive in VR.

3.2 Semantic Embedding and Clustering

To transform the preprocessed video samples into semantically organized structures, we compute multimodal embeddings that capture both visual and textual information. For each video, we extract visual embeddings from representative keyframes using CLIP's image encoder [10], and textual embeddings from associated metadata (titles, transcripts) by CLIP's text encoder. These embeddings are projected into the same latent space, combined by averaging across keyframes for the visual part, and then averaged with the textual representation to form a single vector per video. This ensures that conceptually related videos are positioned closer in the embedding space, even when their visual appearances differ.

Building on this representation, we employ a Large Language Model (LLM) [6,9] to interpret the clustered embedding space rather than directly modifying it. Specifically, LLM takes text summaries of each cluster, generated from video metadata and representative frames, and assigns them understandable topic labels, such as "natural scenery" or "international events". The LLM thus acts as an inference layer, converting raw numerical representations into semantic concepts that users can easily understand and interact with.

To organize the data at scale, we apply hierarchical clustering to the enriched embedding vectors. This process creates higher-level groupings, which we call "video universes". Each universe represents a coherent topic domain, acting as a structured entry point to the entire dataset. This allows users to explore content in universes rather than having to browse thousands of individual videos, making the browsing process more organized, scalable, and cognitively less taxing.

3.3 360° Tapestry Generation

Within each thematic "universe," we construct a 360° panorama offline, combining visual elements from multiple videos into a single immersive composite. This process begins with frame selection: from videos within the same theme, we extract a subset of representative keyframes that reflect the content diversity in that universe. The frames are then projected onto a spherical surface, with the spatial layout optimized to balance coverage and minimize image duplication, drawing inspiration from photo-splicing techniques [25].

To enhance continuity, adjacent frames are gently blended to blur the boundaries, creating a smooth transition across the entire panorama. In addition to still frames, we also insert short clips that repeat local areas, conveying not only image information but also motion signals, creating a more vivid visual effect.

The result is a seamless 360° picture, in which each thematic universe is transformed into an explorable environment. Instead of having to browse each video sequentially, users can navigate in a richly connected space, easily grasping the overview and relationships between videos at a glance.

3.4 Interactive VR-Based Exploration and Retrieval

In the final stage of the system, users can interact with and explore the panorama within the Meta Quest VR headset, transforming the static video collection into a vivid, easy-to-navigate environment. The interactive design prioritizes intuitiveness, allowing users to freely move between overview and detail. By simply turning their head, users can observe the 360° picture, gliding through clusters of topics in a short time. The zoom mechanism allows flexible adjustment of image density, zooming out to see an overview of a universe and zooming in to see smaller details and individual video regions.

Direct interaction is supported through point-and-select gestures, allowing users to highlight an area of interest and immediately access the corresponding original video or clip. In addition to spatial navigation, users can filter by universe and seamlessly switch between topic clusters to focus on specific areas of interest (e.g., sports, nature, international events).

This combination of vivid imagery and semantic structure allows users to get a global overview and retrieve specific content accurately quickly. Such a design is especially effective in competitive Video Browser Showdown (VBS) situations, where speed, accuracy, and intuitive interaction are critical to success.

4 Conclusion

In this paper, we present TapesVRy, a novel system for exploring and retrieving large-scale video collections through immersive 360-degree panoramic images. By integrating keyframe extraction, multimodal clustering, and panoramic carpet imaging, TapesVRy enables users to quickly explore large amounts of videos while improving navigation and retrieval. Unlike traditional approaches, it transforms a passive experience into an active exploration experience.

Our main contributions include (i) a novel 360-degree carpet image representation for video browsing, (ii) the use of multimodal AI for automatic thematic clustering, and (iii) an interactive interface that provides zoom-based density control, seamless navigation, and direct video access.

In the future, we aim to improve our approach by incorporating temporal dynamics, adaptive personalized exploration based on user behavior, and optimization for real-time interaction in VR. We also plan to conduct user studies to evaluate the benefits of immersive exploration compared to traditional video retrieval methods.

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